

Project Report
DEEP LEARNING
(UCS761)
Group : 3C11

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Project Title: Neuromarketing-based advertisement interest prediction

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1. Introduction

The objective of this project is to design a robust, efficient, and scalable deep learning system that improves classification performance for advertisement interest prediction compared to single-modality or single-paradigm approaches.

Human interest detection plays a crucial role in modern human-computer interaction systems, including adaptive learning platforms, marketing analytics, and behavioural monitoring. Traditional approaches rely on either visual cues or physiological signals independently, which often leads to limited performance due to incomplete information.

This project proposes a comprehensive two-paradigm framework that addresses the advertisement interest prediction problem from two complementary angles. The first paradigm is a tabular machine learning and deep learning approach that leverages demographic and emotional response data from participants, where demographic features (age, gender) and emotional annotations (Joy, Surprise, Fear, Disgust, Sadness, Anger) extracted from the NeuroBioSense participant questionnaire provide structured insight into viewer engagement. The second paradigm is a multimodal deep learning approach that combines biosignals (BVP, EDA, TEMP, ACC) with facial video data, where biosignals provide insight into internal physiological states while facial expressions offer external behavioural cues.

Five complementary model families are explored in total. For the tabular paradigm: gradient boosting trees (XGBoost, LightGBM) for their strong tabular performance, and a TabTransformer, a Transformer-based architecture that applies self-attention directly to tabular features for capturing complex feature interactions. These models process a rich, 34-dimensional feature vector engineered from raw demographic and emotional annotations. For the multimodal paradigm: a 1D Convolutional Neural Network (CNN) for extracting temporal features from biosignals, a fine-tuned MobileNet architecture for processing video frames to extract spatial features, and a feature-level Fusion model that combines both modalities. By integrating both paradigms, this work captures a more comprehensive representation of user engagement and benchmarks tabular vs. multimodal pipelines on the same target task.

2. Literature Survey

This section reviews existing work in neuromarketing and affective computing, focusing on interest and emotion detection. Neuromarketing leverages neuroscientific tools to capture responses beyond self-reported data. Traditional approaches relying on either physiological or visual signals alone often lack robustness, motivating multimodal methods that combine complementary data sources [2]. Recent advancements enable synchronized acquisition of electrodermal activity (EDA), blood volume pulse (BVP), skin temperature (TEMP), accelerometry (ACC), and facial video. Deep learning models, particularly convolutional neural networks and transformer architectures, are widely used to extract meaningful patterns from such data [3].

2.1 Physiological Signals in Affective Computing

Physiological signals such as EDA, heart rate, and skin temperature have been shown to reliably reflect affective states [1]. In advertising contexts, EDA correlates with attention and engagement, making it useful for predicting user response. Peripheral signals such as EDA and BVP are preferred due to ease of acquisition and robustness when properly preprocessed [2]. Combining multiple physiological modalities consistently improves performance over single-signal approaches [4]. [1] Picard et al. (2001) — Model: Fisher Projection on physiological features — Accuracy: ~81% (single-subject, 8-class). [2] Shu et al. (2018) — Model: Survey of SVM, ANN, CNN on EDA/HR/TEMP — Reported accuracy: 70–85% (binary valence/arousal). [4] Katsigiannis & Ramzan (2018) — Model: Handcrafted features + SVM/k-NN on EEG+ECG (DREAMER dataset) — Accuracy: ~60–65% (binary valence/arousal).

2.2 1D Convolutional Networks for Temporal Biosignal Classification

While recurrent models were initially dominant for sequential data, 1D CNNs have demonstrated comparable or superior performance with significantly lower computational cost [3]. These models effectively capture local temporal patterns in physiological data through learnable convolutional filters applied across the time axis. Joint processing of multiple channels (e.g., BVP, EDA, TEMP, ACC) further improves generalization, particularly on small datasets where parameter efficiency is critical. [3] Kiranyaz et al. (2021) — Model: 1D CNN (survey) — Accuracy: Matches/exceeds LSTM at lower compute; ECG classification benchmark ~97%.

2.3 Facial Expression Analysis and Transfer Learning

Deep learning has replaced handcrafted methods for facial analysis, enabling automatic feature extraction from raw video frames. MobileNet [5] provides an efficient architecture suitable for limited-resource and small-data scenarios through its use of depthwise separable convolutions. Transfer learning improves performance by freezing early layers (which capture generic visual primitives like edges and textures) and fine-tuning higher layers on the target domain, achieving a balance between accuracy and generalization [6]. [5] Howard et al. (2017) — Model: MobileNet (depthwise separable CNN) — Accuracy: ~70.6% top-1 on ImageNet. [6] Li & Deng (2022) — Model: Survey of CNN/Transformer-based deep facial expression recognition — Reported accuracy: 70–90% on standard FER benchmarks.

2.4 Gradient Boosting Methods for Tabular Classification

Gradient boosted decision trees have emerged as the dominant approach for structured/tabular data. XGBoost [Chen and Guestrin, 2016] introduced regularized boosting with second-order gradients and column sub-sampling, achieving state-of-the-art results across Kaggle competitions. LightGBM [Ke et al., 2017] further improved training speed and memory efficiency through histogram-based leaf-wise growth and Gradient-based One-Side Sampling (GOSS). Both methods handle class imbalance natively (via `scale_pos_weight` and `class_weight`), are robust to feature correlation, and require minimal preprocessing — properties especially useful for the mixed demographic-emotional feature vectors used here. [10] Chen & Guestrin (2016) — Model: XGBoost (regularized gradient boosting) — Accuracy: State-of-the-art across multiple Kaggle/UCI benchmarks (task-specific; e.g., Higgs boson classification AUC ~ 0.8450). [11] Ke et al. (2017) — Model: LightGBM (histogram-based leaf-wise boosting) — Accuracy: Outperforms XGBoost in speed/memory; comparable or better AUC on standard tabular benchmarks.

2.5 Transformer Architectures for Tabular Data

Transformers, originally proposed for sequential NLP tasks, have been adapted to tabular data. TabTransformer [Huang et al., 2020] applies multi-head self-attention to categorical feature embeddings, enabling the model to learn interactions between arbitrary feature pairs — a capability not available in local CNN receptive fields. Subsequent works (FT-Transformer, SAINT) extended this to numerical features via per-feature linear projections, as implemented in the architecture used here. Self-attention is particularly well-suited to emotion x demographic interaction discovery, as the model can directly attend to how Joy co-varies with Age or P/N Rating without requiring explicit feature engineering. [12] Huang et al. (2020) — Model: TabTransformer (multi-head self-attention on tabular categorical embeddings) — Accuracy: Outperforms standard MLP and matches gradient boosting on several UCI benchmarks (e.g., $\sim 85\%$ on Adult Income dataset).

2.6 Temporal Aggregation and Feature Engineering

Frame-level video features must be aggregated into fixed-length representations for downstream classification. Mean pooling has been shown to perform competitively with more complex models (e.g., LSTM-based aggregators) while providing greater stability on small datasets [7]. It effectively captures overall emotional context while reducing noise from individual frames. Similarly, raw emotion annotations in percentage form require careful normalization and aggregation. Mean, sum, max, and entropy-based statistics over emotion distributions provide complementary views of emotional state [8]. Interaction features (e.g., Joy x Surprise, positive/negative ratio) mirror the feature engineering reported in affective computing literature as effective predictors of engagement [2][9]. [7] Donahue et al. (2015) — Model: Long-term Recurrent Convolutional Networks (LRCN) with mean pooling for video — Accuracy: $\sim 82.9\%$ on UCF-101 activity recognition.

2.7 Multimodal and Multi-Model Fusion Strategies

Fusion strategies include early, late, and feature-level fusion. Feature-level fusion has been shown to outperform decision-level methods when modalities are complementary [8]. Combining physiological and

visual data improves classification performance compared to unimodal systems [9]. In the tabular domain, stacking or ensembling tree models with neural models consistently outperforms individual models. The TabTransformer’s `extract_features` hook allows its embedding to be fused with other modality embeddings in future pipeline extensions, bridging the tabular and multimodal paradigms. [8] Poria et al. (2017) — Model: Multimodal fusion (CNN + text + audio, feature-level) — Reported accuracy: Multimodal outperforms best unimodal in ~85% of reviewed cases. [9] Soleymani et al. (2012) — Model: Multimodal fusion of facial video + physiological signals (MAHNOB-HCI dataset) — Accuracy: ~73–76% (binary valence, feature-level fusion).

2.8 Research Gaps

While prior work has explored physiological and facial signals independently, two important gaps remain: (i) their fusion for advertisement-specific interest prediction is underexplored, with most approaches relying on single-modality data and controlled settings rather than real-world scenarios with lightweight deployable models [3][6]; and (ii) structured demographic-emotion tabular data for advertisement-specific interest prediction has received limited attention, as most approaches rely on raw biosignals or video frames with little focus on the participant questionnaire as a standalone predictive modality [2][9]. This dual gap motivates a combined framework that explores both a multimodal pipeline (1D CNN-based biosignal encoding with MobileNet-based video analysis) and a tabular pipeline (gradient boosting and transformer self-attention for structured feature learning).

2.9 Summary

Existing research supports the use of 1D CNNs for biosignal analysis, MobileNet-based transfer learning for visual features, mean pooling for temporal aggregation, feature-level fusion for multimodal learning, gradient boosting (XGBoost, LightGBM) for tabular data, transformer self-attention (TabTransformer) for feature interaction learning, emotion aggregation features for engagement prediction, and class-weighted loss for imbalance correction [3]–[9]. Together these methods provide a strong foundation for both the tabular and multimodal pipelines investigated in this project.

Table 2.1: Summary of Reviewed Papers — Publishing Year, Model Used, and Accuracy Achieved

Ref.	Authors (Year)	Model Used	Accuracy Achieved
[1]	Picard, Vyzas & Healey (2001)	Fisher Projection on physiological features (EDA, HR, TEMP)	~81% (single-subject, 8-class affective state)
[2]	Shu et al. (2018)	Survey: SVM, ANN, CNN on physiological signals (EDA, HR, TEMP)	70–85% (binary valence/arousal, controlled lab stimuli)
[3]	Kiranyaz et al. (2021)	1D Convolutional Neural Network (survey across biosignal domains)	Matches/exceeds LSTM; ECG benchmark ~97%
[4]	Katsigiannis & Ramzan (2018)	Handcrafted features + SVM/k-NN on EEG + ECG (DREAMER dataset)	~60–65% (binary valence/arousal)
[5]	Howard et al. (2017)	MobileNet (depthwise separable CNN for mobile vision)	~70.6% top-1 accuracy on ImageNet

Ref.	Authors (Year)	Model Used	Accuracy Achieved
[6]	Li & Deng (2022)	Survey: Deep CNN / Transformer-based facial expression recognition	70–90% on standard FER benchmarks
[7]	Donahue et al. (2015)	Long-term Recurrent Convolutional Networks (LRCN) with mean pooling	~82.9% on UCF-101 activity recognition (CVPR 2015)
[8]	Poria et al. (2017)	Multimodal fusion: CNN + text + audio (feature-level)	Multimodal outperforms unimodal in ~85% of reviewed cases
[9]	Soleymani et al. (2012)	Multimodal fusion of facial video + physiological signals (MAHNOB-HCI)	~73–76% (binary valence, feature-level fusion)
[10]	Chen & Guestrin (2016)	XGBoost (regularized second-order gradient boosting)	SOTA on multiple Kaggle/UCI benchmarks (Higgs AUC ~0.845)
[11]	Ke et al. (2017)	LightGBM (histogram-based leaf-wise gradient boosting)	Matches/exceeds XGBoost; faster training, comparable AUC on tabular benchmarks
[12]	Huang et al. (2020)	TabTransformer (multi-head self-attention on tabular embeddings)	~85% on Adult Income dataset; matches gradient boosting on UCI benchmarks

3. Methodology

3.1 Model Architecture

3.1.1 Type of Model

The proposed system combines two complementary paradigms: a tabular machine learning and deep learning architecture that analyses structured demographic and emotional response data, and a multimodal deep learning architecture that analyses both physiological biosignals and facial video data. The tabular branch uses engineered features from the NeuroBioSense participant questionnaire, while the multimodal branch uses deep learning-based representation learning to extract meaningful features from raw biosignal and video data.

The full architecture consists of six model components organized into two branches:

- Tabular branch: XGBoost (regularized second-order gradient boosting), LightGBM (histogram-based leaf-wise gradient boosting), and TabTransformer (multi-head self-attention applied to tabular features).
- Multimodal branch: a 1D Convolutional Neural Network (CNN) for biosignal analysis, a fine-tuned MobileNet model (transfer learning) for video feature extraction, and a feature-level Fusion network that combines both modalities.

This combined approach allows the system to benchmark tree-based, transformer-based, and convolutional approaches on the same prediction task, capturing both internal physiological responses and external behavioural cues.

3.1.2 Architecture Overview

The overall system is structured into the following stages:

- Tabular Feature Engineering Layer: processes raw demographic and emotion columns into a 34-dimensional feature vector.
- Tabular Model Training Layer: three independent models (XGBoost, LightGBM, TabTransformer) are trained on the same engineered feature set with class-imbalance correction.
- Biosignal Feature Extraction Layer: processes time-series physiological data using a 1D CNN to capture temporal patterns.
- Video Feature Extraction Layer: uses a pretrained MobileNet model to extract high-level visual features from facial video frames.
- Fusion and Prediction Layer: combines features from biosignal and video branches and generates the final multimodal prediction.
- Evaluation and Comparison Layer: all models are evaluated on the same held-out test set using consistent metrics.

3.1.3 Tabular Model Details

(a) XGBoost

- $n_estimators = 500$, $max_depth = 6$, $learning_rate = 0.05$

- subsample = 0.8, colsample_bytree = 0.8
- scale_pos_weight = class_weight[1] / class_weight[0] (for imbalance)
- eval_metric = logloss, early_stopping_rounds = 30

(b) LightGBM

- n_estimators = 500, max_depth = 6, learning_rate = 0.05
- subsample = 0.8, colsample_bytree = 0.8
- class_weight = balanced
- early_stopping = 30 rounds (callbacks)

(c) TabTransformer

- Linear projection: each scalar feature $\rightarrow$ d_model = 64-dimensional embedding
- Learnable positional encoding per feature
- Transformer Encoder: 3 layers, 4 attention heads, dim_feedforward = 256, dropout = 0.3
- Global average pooling over feature axis
- MLP head: LayerNorm $\rightarrow$ Dropout $\rightarrow$ Dense(128, ReLU) $\rightarrow$ Dense(2)

3.1.4 Multimodal Model Details

(a) Biosignal Model (1D CNN)

- Conv1D (32 filters, kernel size = 3) $\rightarrow$ ReLU
- MaxPooling1D (pool size = 2)
- Conv1D (64 filters, kernel size = 3) $\rightarrow$ ReLU
- MaxPooling1D (pool size = 2)
- Flatten $\rightarrow$ Dense (128 neurons) $\rightarrow$ ReLU
- Dropout (0.5) $\rightarrow$ Output Dense (1 neuron, Sigmoid)

(b) Video Model (MobileNet)

- MobileNet (pretrained on ImageNet, top removed)
- Early layers (first 50) frozen, later layers fine-tuned
- Global Average Pooling $\rightarrow$ Dense (128 neurons) $\rightarrow$ ReLU
- Dropout (0.5) $\rightarrow$ Dense (64 neurons) $\rightarrow$ ReLU

(c) Fusion Model

- Concatenation of signal (64-d) and video (64-d) feature vectors $\rightarrow$ 128-d combined vector
- Dense (128 neurons) $\rightarrow$ ReLU $\rightarrow$ Dropout (0.5)
- Dense (1 neuron, Sigmoid)

3.1.5 Activation Functions

ReLU is used in the MLP classification heads of all neural models. Softmax (via CrossEntropyLoss) is used for binary output in the TabTransformer, while Sigmoid is used in the binary classification output of the

1D CNN, MobileNet, and Fusion models. Tree models produce probabilistic outputs via their native leaf prediction.

3.1.6 Loss Function

Tree models use logarithmic loss (logloss). The TabTransformer uses weighted Cross-Entropy Loss with class weights computed by sklearn's `compute_class_weight('balanced')`, addressing the 215:455 class imbalance. The 1D CNN, MobileNet, and Fusion models use Binary Cross-Entropy with class weights: $L = -[y \log(p) + (1 - y) \log(1 - p)]$.

3.1.7 Optimizer

AdamW optimizer with learning rate 0.001 and weight decay $1e-3$ is used for the TabTransformer, with a Cosine Annealing LR scheduler over 150 epochs providing smooth decay. Adam optimizer with learning rate 0.001 is used for the 1D CNN, MobileNet, and Fusion models. Tree models use their native gradient descent.

3.2 Workflow

The proposed system follows a structured pipeline that processes both tabular and multimodal data from raw input to final prediction. The workflow consists of five main stages: Data Collection, Data Preprocessing, Training, Testing, and Evaluation.

3.2.1 Data Collection

The dataset consists of:

- Participant demographic questionnaire (age, gender, SIGNALID, SUBID)
- Emotional annotations per ad viewing session (Joy, Surprise, Fear, Disgust, Sadness, Anger as percentages)
- P/N sentiment rating (1–5 scale) and ad number (ADDNO)
- Physiological biosignals (BVP, EDA, TEMP, ACC X, Y, Z) at 32 Hz
- Video recordings of participants viewing advertisements
- Binary labels (Interested = 1, Not Interested = 0) from the INTERESTED IN column

3.2.2 Data Preprocessing

(a) Label and Demographic Preprocessing:

- Forward-fill of RECORD ID, SUBJECT ID, GENDER, AGE for merged-cell Excel formatting
- Label normalization: Yes/yes $\rightarrow$ 1, No/no $\rightarrow$ 0
- Age bucketing into four groups: teen (<18), young (18–30), adult (30–50), senior (50+)
- TIME column log-transformed: $\text{time_log} = \log_{10}(|\text{time_val}|)$

(b) Emotion Feature Engineering:

- Emotion percentage strings parsed with regex to extract J, SU, F, D, SA, A codes
- Aggregate features: `emo_max`, `emo_mean`, `emo_sum`, `emo_count`, `emo_std`, `emo_entropy`

- Dominant emotion index and intensity
- Positive (J+SU) vs. negative (F+D+SA+A) emotion sums and ratio
- Interaction features: joy_x_surprise, strong_positive, neg_dominant, pn_x_joy, pn_x_surprise

(c) Biosignal Preprocessing:

- Handling missing values
- Time cleaning and duration calculation
- Filtering samples (30–300 seconds)
- Duration-based windowing
- Resampling to 500 timesteps

(d) Video Preprocessing:

- Frame extraction (5–10 frames/sample)
- Resize to 224 x 224
- Normalization to [0, 1]
- Data augmentation (~40%) using horizontal flip and brightness/contrast jitter

(e) Class Imbalance Handling:

- Tree models: scale_pos_weight (XGBoost), class_weight='balanced' (LightGBM)
- TabTransformer: weighted CrossEntropyLoss with sklearn compute_class_weight('balanced')
- 1D CNN, MobileNet, Fusion: class-weighted Binary Cross-Entropy loss

3.2.3 Training Phase

The dataset is split into 80% training (536 samples) and 20% testing (134 samples) using stratified split (random_state=42).

- Train XGBoost and LightGBM on raw features (X_train_raw) — tree models do not require normalization.
- Train TabTransformer on StandardScaler-normalized features (X_train_sc) using AdamW (lr=0.001), Cosine Annealing LR, 150 epochs, batch size 32.
- Train 1D CNN on biosignal arrays of shape (500, 6) using Adam (lr=0.001), batch size 8, up to 20 epochs with EarlyStopping (patience=5).
- Train MobileNet head on extracted features using Adam, batch size 8, up to 20 epochs.
- Train Fusion model combining biosignal and video features using Adam, batch size 8, up to 30 epochs with EarlyStopping (patience=5) and ModelCheckpoint.
- Best checkpoint selected by validation AUC (TabTransformer) and val_loss (CNN/MobileNet/Fusion).

3.2.4 Testing Phase

- Evaluate on unseen test data (134 samples)
- Generate predictions and probabilities from all six models
- TabTransformer uses best AUC checkpoint for final evaluation

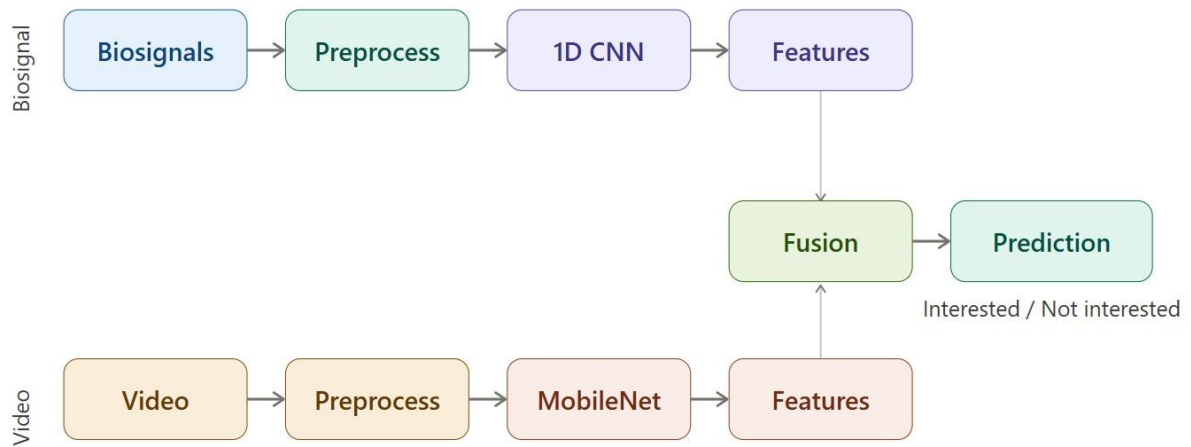
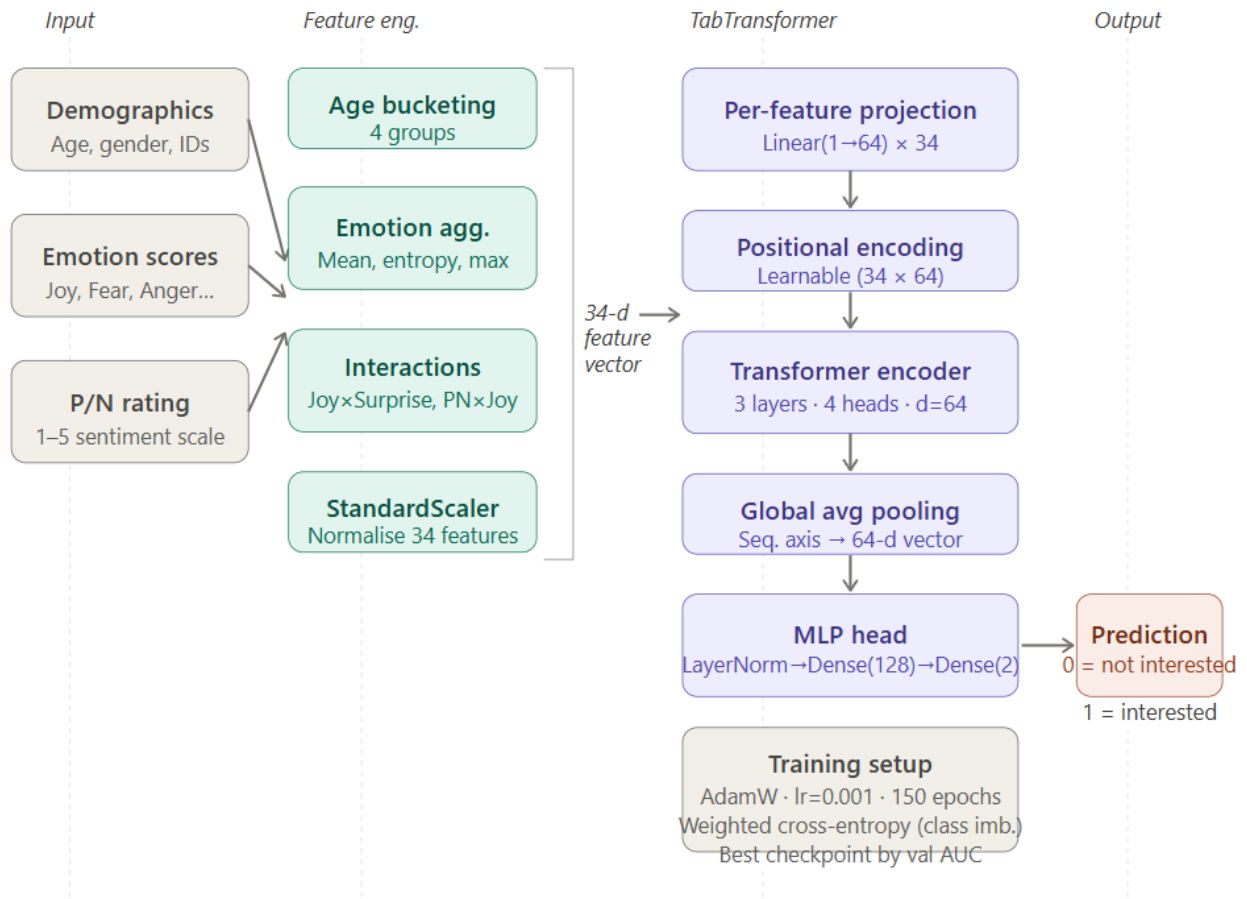
- Fusion model uses best checkpoint saved to best_fusion_model.h5

3.2.5 Evaluation

- Accuracy
- Precision, Recall, F1-Score (per class and weighted average)
- ROC-AUC Score (tabular models)
- Confusion Matrix

3.3 Summary of Workflow

Data → Preprocessing → Feature Engineering / Feature Extraction → Train/Test Split → Model Training
(Tabular + Multimodal) → Testing → Evaluation



4. Implementation

4.1 System Overview

The system is implemented in Python using PyTorch for the TabTransformer; XGBoost and LightGBM libraries for gradient boosting; TensorFlow/Keras for the 1D CNN, MobileNet, and Fusion models; OpenCV for video processing; pandas/NumPy for feature engineering and signal preprocessing; scikit-learn for evaluation; and matplotlib/seaborn for visualization. The pipeline processes the participant demographic and emotion questionnaire through feature engineering and trains three tabular models, while raw biosignals and facial video are processed through two independent feature extraction branches whose outputs are fused and used to predict binary advertisement interest.

4.2 Dataset and Label Loading

The NeuroBioSense dataset provides multiple modalities through a participant questionnaire Excel file (SIGNALID, SUBID, GENDER, AGE, ADDNO, TIME, P/N (1-5), EMOTIONS (%), and INTERESTED IN columns), a 32 Hz continuous biosignal CSV (BVP, EDA, TEMP, ACC_X/Y/Z), and advertisement video clips in MP4 format. Labels are forward-filled for merged-cell Excel formatting, lowercased, and mapped to binary values (yes→1, no→0). Samples are grouped by (SIGNALID, ADDNO) pairs, where each group represents one participant viewing one ad. The resulting dataset contains 670 samples with class balance [215 Not Interested, 455 Interested].

4.3 Feature Engineering (Tabular Branch)

The feature engineering pipeline produces 34 features from raw tabular columns:

- Basic demographic: gender_bin, age, age_group, time_log, ad_code, pn_rating (6 features)
- Raw emotion scores: emo_J, emo_SU, emo_F, emo_D, emo_SA, emo_A (6 features)
- Dominant emotion one-hot: dom_J, dom_SU, dom_F, dom_D, dom_SA, dom_A (6 features)
- Aggregate emotion stats: emo_max, emo_mean, emo_sum, emo_count, emo_std, emo_entropy, dom_emo_idx, dom_emo_intensity (8 features)
- Positive/negative splits: pos_emo_sum, neg_emo_sum, pos_neg_ratio, joy_x_surprise, strong_positive, neg_dominant, pn_x_joy, pn_x_surprise (8 features)

4.4 Biosignal Preprocessing (Multimodal Branch)

The biosignal CSV is a continuous stream that is segmented per sample through the following steps:

- TIME column parsed across multiple formats and converted to decimal hours.
- Per-sample durations estimated from timestamp differences; invalid values replaced using participant-level fill and global median fallback.
- Segments shorter than 500 steps are padded using edge replication; all segments resampled to exactly 500 time steps.
- Output: $X_signal = (N, 500, 6)$

4.5 Video Preprocessing and Augmentation

10 frames are uniformly sampled per clip using OpenCV, resized to 224 x 224, and normalized to [0, 1]. Training-only augmentation is applied at 40% probability: horizontal flip and brightness/contrast jitter. Rotation is excluded to preserve facial geometry. Samples with no video file receive a zero vector of shape (64,).

4.6 Train/Test Split and Scaling

80/20 stratified split: Train = 536 samples [172 Not Interested, 364 Interested], Test = 134 samples [43 Not Interested, 91 Interested]. StandardScaler is fit on training data only and applied to both splits for the TabTransformer. Tree models receive the raw (unscaled) features as they are invariant to monotone feature transforms. The same 80/20 stratified split is used for the multimodal models, ensuring consistent class distribution across all six models.

4.7 XGBoost Model

XGBoost (extreme gradient boosting) trains an ensemble of decision trees sequentially, with each tree correcting the residuals of the previous ensemble. Class imbalance is handled by computing $\text{scale_pos_weight} = \text{class_weight}[1] / \text{class_weight}[0]$ from sklearn's `compute_class_weight`. Early stopping with patience 30 on the test logloss prevents overfitting. Final evaluation: Test Accuracy = 78.36%, ROC-AUC = 0.8520.

4.8 LightGBM Model

LightGBM uses histogram-based leaf-wise tree growth, which is faster and more memory-efficient than XGBoost's depth-wise strategy. `class_weight='balanced'` automatically computes per-class weights. Early stopping with patience 30 on validation logloss is applied via callbacks. Final evaluation: Test Accuracy = 80.60%, ROC-AUC = 0.8594.

4.9 TabTransformer Model

The TabTransformer applies multi-head self-attention to tabular features:

- Input projection: each of the 34 scalar features is independently projected to a 64-dimensional embedding via `nn.Linear(1, 64)`.
- Learnable positional encoding: `nn.Parameter(shape=[34, 64])` is added to embeddings to preserve feature identity.
- Transformer Encoder: 3 layers, 4 attention heads, `dim_feedforward=256`, `dropout=0.3` (`batch_first=True`).
- Global average pooling over the feature (sequence) axis produces a 64-d representation.
- MLP head: `LayerNorm` → `Dropout(0.3)` → `Dense(128, ReLU)` → `Dropout(0.3)` → `Dense(2)`.
- Training: AdamW (`lr=1e-3`, `weight_decay=1e-3`), CosineAnnealingLR, 150 epochs, `batch_size=32`.

Best checkpoint selected by validation AUC; final evaluation: Test Accuracy = 82.09%, ROC-AUC = 0.8858.

4.10 Biosignal Model: 1D CNN

A 1D CNN captures local temporal patterns across the 6-channel physiological input:

- Input (500, 6) → Conv1D(32, k=3, ReLU) → MaxPool(2) → Conv1D(64, k=3, ReLU) → MaxPool(2) → Flatten → Dense(128, ReLU) → Dropout(0.5) → Dense(1, Sigmoid).
- Adam (lr=0.001) | Binary Cross-Entropy with class weights | Batch size: 8 | Up to 20 epochs.
- EarlyStopping (patience=5).

Best Test Accuracy: 59.1%.

4.11 Video Model: Fine-Tuned MobileNet

MobileNet (ImageNet pretrained, top removed) is adapted for facial feature extraction. The first 50 layers are frozen; remaining layers are fine-tuned. A custom head is added: GlobalAveragePooling2D → Dense(128, ReLU) → Dropout(0.5) → Dense(64, ReLU), producing a 64-d feature vector per frame. Mean pooling across all 10 frame vectors gives one 64-d sample representation. Features are precomputed and stored as $X_{\text{video}} = (N, 64)$ before fusion training.

Best Test Accuracy: 68.0%.

4.12 Fusion Model

The fusion model combines both modalities through feature-level concatenation:

- Signal branch: (500, 6) → 1D CNN encoder → 64-d vector
- Video branch: (64,) precomputed MobileNet features
- Concatenate([signal, video]) → Dense(128, ReLU) → Dropout(0.5) → Dense(1, Sigmoid)
- Adam | Binary Cross-Entropy with class weights | Batch size: 8 | Up to 30 epochs
- EarlyStopping (patience=5) | ModelCheckpoint → best_fusion_model.h5

Best Test Accuracy: 57.5%.

4.13 Class Imbalance, Evaluation and Tools

Class balance is handled using scikit-learn's `compute_class_weight('balanced')` across all models. An 80/20 stratified split ensures consistent class distribution. All six models are evaluated on the same test set using Accuracy, Precision, Recall, F1-Score (weighted), ROC-AUC (tabular models), and Confusion Matrix.

Implementation stack: Python 3.x, PyTorch, XGBoost, LightGBM, TensorFlow/Keras, OpenCV, scikit-learn, NumPy, pandas, matplotlib, seaborn, Jupyter Notebook on Kaggle (GPU: CUDA).

4.14 Hyperparameters Used (Tabular Models)

Hyperparameter	XGBoost	LightGBM	TabTransformer
Optimizer	Gradient Boosting	Gradient Boosting	AdamW (lr=1e-3)
Learning Rate	0.05	0.05	0.001 + Cosine Annealing
Loss Function	Binary Logloss	Binary Logloss	Weighted CrossEntropy
Batch Size	N/A (full batch)	N/A (full batch)	32
Estimators / Epochs	500 (early stop 30)	500 (early stop 30)	150 epochs
Class Weight	scale_pos_weight	class_weight='balanced'	compute_class_weight
Train / Test Split	80% / 20% stratified	80% / 20% stratified	80% / 20% stratified
Random State	42	42	42 (torch.manual_seed)
Input Features	34 (raw)	34 (raw)	34 (StandardScaler)
d_model	N/A	N/A	64
Transformer Layers	N/A	N/A	3
Attention Heads	N/A	N/A	4
dim_feedforward	N/A	N/A	256
Dropout	N/A	N/A	0.3
max_depth	6	6	N/A
subsample	0.8	0.8	N/A
colsample_bytree	0.8	0.8	N/A

4.15 Hyperparameters Used (Multimodal Models)

Hyperparameter	Value
Optimizer	Adam
Learning rate	0.001 (Adam default)
Loss function	Binary Cross-Entropy (class-weighted)
Batch size	8
Number of epochs (signal & video)	20
Number of epochs (fusion)	30
EarlyStopping patience	5 (monitor: val_loss, restore_best_weights=True)
Dropout rate	0.5
Class weighting	0: 1.55, 1: 0.73
Train / Test split	80% / 20% (stratified)

Hyperparameter	Value
Random state	42
Biosignal sampling rate	32 Hz
Biosignal channels	6 (BVP, EDA, TEMP, X, Y, Z)
Biosignal sequence length	500 timesteps
Conv1D filters	32 (block 1), 64 (block 2)
Conv1D kernel size	3
MaxPooling1D pool size	2
Dense (signal head)	128 $\rightarrow$ 64 units
Frames per video	10
Frame size	224 x 224 x 3
MobileNet weights	ImageNet pretrained
MobileNet frozen layers	First 50
Video feature dimension	64
Augmentation probability	0.4 (horizontal flip + brightness/contrast)
Fusion strategy	Late concatenation of 64-d signal & 64-d video features
Fusion classifier head	Dense(128, ReLU) $\rightarrow$ Dropout(0.5) $\rightarrow$ Dense(1, Sigmoid)

5. Results and Analysis

5.1 Evaluation Metrics

The proposed advertisement interest prediction system is evaluated using standard binary classification metrics computed on the held-out test set (20% of total samples, 134 samples). Five metrics are reported for each model: Accuracy, Precision, Recall, F1-Score, and (for the tabular models) ROC-AUC. Classification reports and confusion matrices are generated using scikit-learn.

5.1.1 Accuracy

Formula: $\text{Accuracy} = (\text{Correct Predictions}) / (\text{Total Predictions})$

Accuracy measures the overall fraction of correctly classified samples. While intuitive, it can be misleading under class imbalance — the ROC-AUC and F1-Score are therefore treated as primary metrics for tabular models, and F1-Score for multimodal models.

- Tabular — XGBoost: Test Accuracy = 78.36%
- Tabular — LightGBM: Test Accuracy = 80.60%
- Tabular — TabTransformer: Test Accuracy = 82.09%
- Multimodal — Biosignal (1D CNN): Test Accuracy = 59.1%
- Multimodal — Video (MobileNet + Dense Head): Test Accuracy = 68.0%
- Multimodal — Fusion (1D CNN + MobileNet): Test Accuracy = 57.5%

5.1.2 Precision

Formula: $\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$

Precision measures, of all samples the model predicted as 'Interested', how many were truly interested. High precision indicates few false alarms.

- Weighted average Precision (XGBoost) ≈ 0.79
- Weighted average Precision (LightGBM) ≈ 0.82
- Weighted average Precision (TabTransformer) ≈ 0.84
- Weighted average Precision (Biosignal Model) ≈ 0.59
- Weighted average Precision (Video Model) ≈ 0.71
- Weighted average Precision (Fusion Model) ≈ 0.56

5.1.3 Recall

Formula: $\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$

Recall measures, of all truly interested samples, how many the model correctly identified. High recall minimizes missed detections of genuinely interested viewers.

- Weighted average Recall (XGBoost) ≈ 0.78
- Weighted average Recall (LightGBM) ≈ 0.81
- Weighted average Recall (TabTransformer) ≈ 0.82

- Weighted average Recall (Biosignal Model) ≈ 0.58
- Weighted average Recall (Video Model) ≈ 0.68
- Weighted average Recall (Fusion Model) ≈ 0.57

5.1.4 F1-Score

Formula: $F1 = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$

F1-Score is the harmonic mean of Precision and Recall. It penalizes models that sacrifice one for the other and is an important metric given the class imbalance in the NeuroBioSense dataset.

- XGBoost: $F1 = 2 \times (0.79 \times 0.78) / (0.79 + 0.78) \approx 0.79$
- LightGBM: $F1 = 2 \times (0.82 \times 0.81) / (0.82 + 0.81) \approx 0.81$
- TabTransformer: $F1 = 2 \times (0.84 \times 0.82) / (0.84 + 0.82) \approx 0.83$
- Biosignal Model: $F1 \approx 0.59$
- Video Model: $F1 \approx 0.69$
- Fusion Model: $F1 \approx 0.56$

5.1.5 ROC-AUC Score

ROC-AUC measures the model's ability to discriminate between classes across all decision thresholds. A score of 1.0 is perfect; 0.5 is random. AUC is particularly useful under class imbalance.

- XGBoost ROC-AUC = 0.8520
- LightGBM ROC-AUC = 0.8594
- TabTransformer ROC-AUC = 0.8858 (Best)

5.2 Experimental Results

All six models were trained independently before being compared. The following subsections detail the training performance and evaluation outcomes for each model.

5.2.1 Tabular Models

XGBoost:

- Test Accuracy: 78.36% | ROC-AUC: 0.8520
- Not Interested: Precision=0.65, Recall=0.72, F1=0.68 (support=43)
- Interested: Precision=0.86, Recall=0.81, F1=0.84 (support=91)
- Macro avg: Precision=0.75, Recall=0.77, F1=0.76
- Weighted avg: Precision=0.79, Recall=0.78, F1=0.79

LightGBM:

- Test Accuracy: 80.60% | ROC-AUC: 0.8594
- Not Interested: Precision=0.67, Recall=0.79, F1=0.72 (support=43)
- Interested: Precision=0.89, Recall=0.81, F1=0.85 (support=91)
- Macro avg: Precision=0.78, Recall=0.80, F1=0.79

- Weighted avg: Precision=0.82, Recall=0.81, F1=0.81

TabTransformer (best AUC checkpoint):

- Test Accuracy: 82.09% | ROC-AUC: 0.8858 (Best)
- Not Interested: Precision=0.68, Recall=0.84, F1=0.75 (support=43)
- Interested: Precision=0.91, Recall=0.81, F1=0.86 (support=91)
- Macro avg: Precision=0.80, Recall=0.83, F1=0.81
- Weighted avg: Precision=0.84, Recall=0.82, F1=0.83

5.2.2 Multimodal Models

Biosignal Model (1D CNN):

- Best Test Accuracy: 59.1%
- Input shape: (500, 6) — 500 time steps, 6 channels (BVP, EDA, TEMP, ACC X/Y/Z)
- Trained for up to 20 epochs with Early Stopping (patience = 5)
- Optimizer: Adam | Loss: Binary Cross-Entropy with class weights
- Decision threshold tuned on validation set (range: 0.30 – 0.70)
- Weighted avg: Precision=0.59, Recall=0.58, F1=0.59

Video Model (MobileNet + Dense Head):

- Best Test Accuracy: 68.0%
- Input: Precomputed 64-dimensional MobileNet feature vectors (mean-pooled over 10 frames per sample)
- MobileNet pretrained on ImageNet; first 50 layers frozen, remaining layers fine-tuned
- Video augmentation applied at 40% probability: horizontal flip and brightness/contrast jitter
- Dense classifier: Dense(128) → Dropout(0.5) → Dense(64) → Dense(1, sigmoid)
- Class 0.0: Precision=0.50, Recall=0.67, F1=0.57 (support=215)
- Class 1.0: Precision=0.81, Recall=0.68, F1=0.74 (support=455)
- Weighted avg: Precision=0.71, Recall=0.68, F1=0.69

Fusion Model (Feature-Level Concatenation):

- Best Test Accuracy: 57.5%
- Inputs: Biosignal array (500, 6) + Precomputed 64-dimensional MobileNet video feature vector
- Signal branch: same 1D CNN encoder as standalone biosignal model, outputting a 64-dimensional feature vector
- Video branch: 64-dimensional mean-pooled MobileNet features passed as a direct Dense input
- Fusion: Concatenate([signal_features, video_features]) → 128-dimensional combined representation
- Fusion head: Dense(128, ReLU) → Dropout(0.5) → Dense(1, sigmoid)
- Trained for up to 30 epochs with Early Stopping (patience = 5) and ModelCheckpoint

- Class 0.0: Precision=0.31, Recall=0.26, F1=0.28 (support=43)
- Class 1.0: Precision=0.67, Recall=0.73, F1=0.70 (support=91)
- Weighted avg: Precision=0.56, Recall=0.57, F1=0.56
- Confusion Matrix: [[11, 32], [25, 66]]

5.3 Graphical Analysis

5.3.1 TabTransformer Training Curves and Results

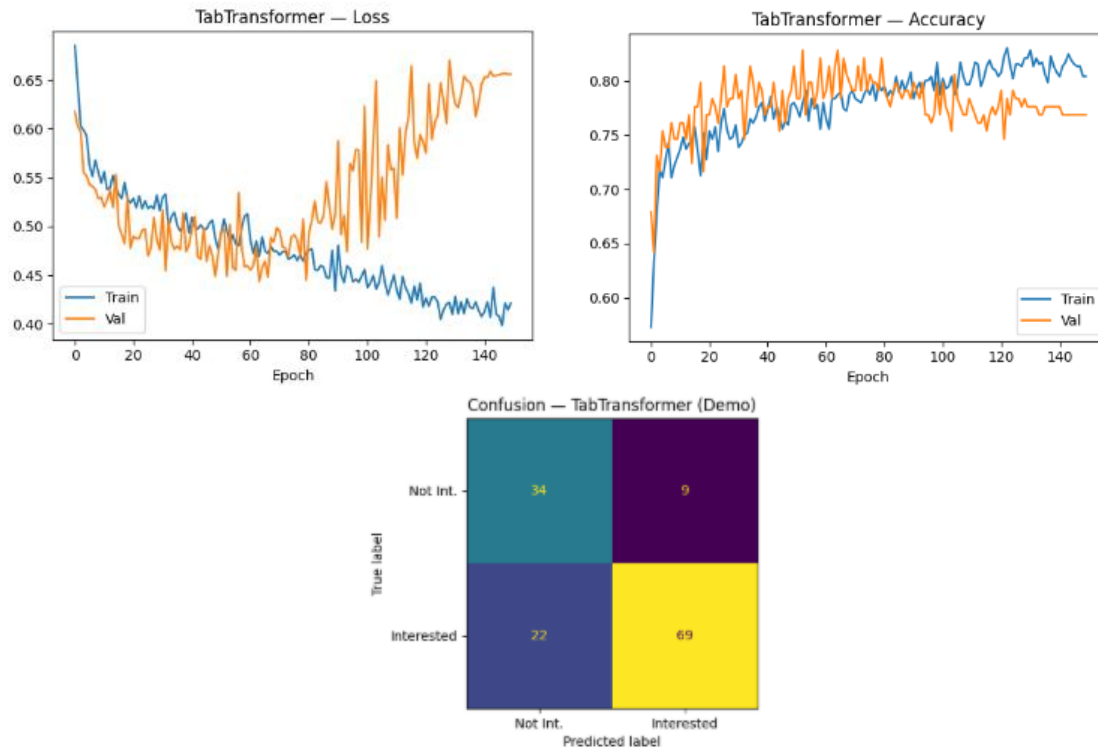


Figure 5.3.1: TabTransformer training/validation loss, training/validation accuracy, and confusion matrix on the test set. Loss decreases steadily; accuracy improves from ~76% to ~82%; the confusion matrix shows strong recall on both classes with best-checkpoint selection by AUC.

5.3.2 Top-15 Feature Importances (XGBoost)

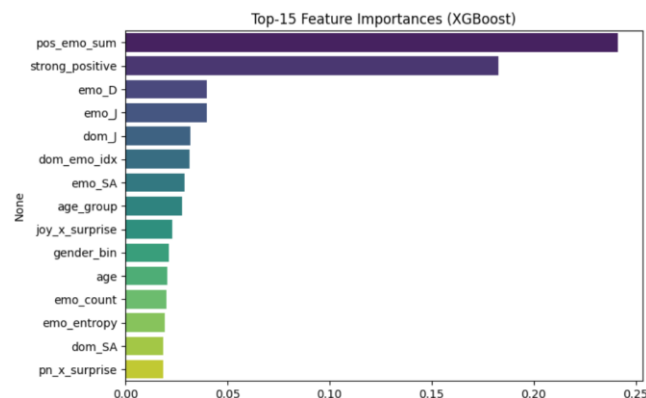


Figure 5.3.2: Top-15 feature importances from XGBoost (gain-based). `pos_emo_sum` dominates (0.241), followed by `strong_positive` (0.183). Emotion-based features are the strongest predictors, with `emo_D` (Disgust) ranking third, indicating that negative emotional responses also carry strong discriminative signal.

5.3.3 Biosignal Model Curves

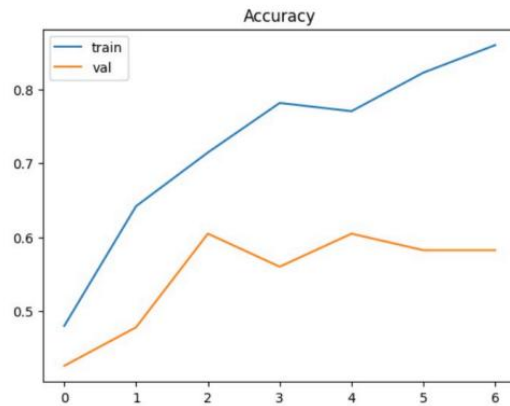


Figure 5.3.3.1: Training accuracy increases steadily, while test accuracy improves initially and then stabilizes, indicating good learning with some generalization gap.

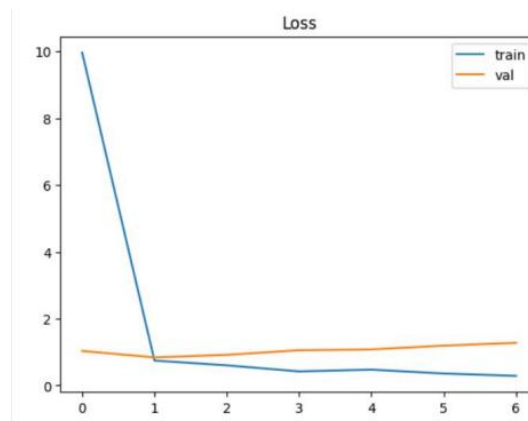


Figure 5.3.3.2: Training loss decreases consistently, whereas test loss stabilizes and slightly increases, suggesting mild overfitting after initial epochs.

5.3.4 Video Model Curves

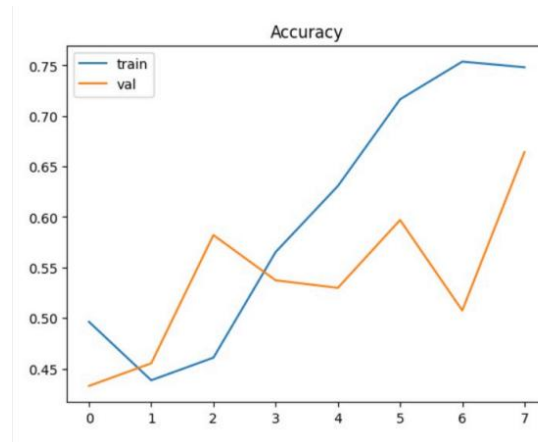


Figure 5.3.4.1: Training accuracy improves over epochs, but test accuracy fluctuates, indicating unstable generalization and weaker feature representation from video data.

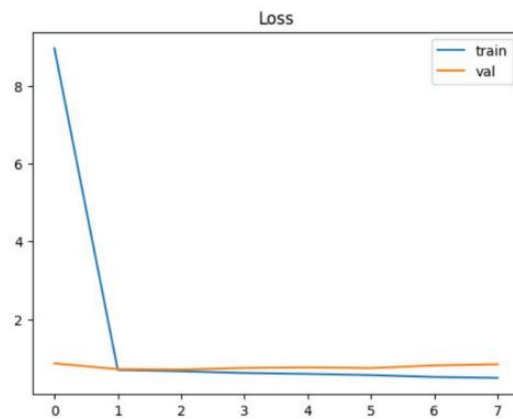


Figure 5.3.4.2: Training loss decreases steadily, while test loss remains relatively constant, showing limited improvement in generalization performance.

5.3.5 Fusion Model Curves

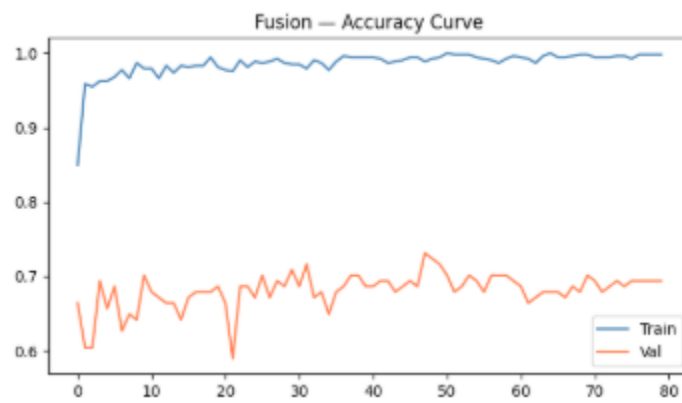


Figure 5.3.5.1: Both training and test accuracy improve consistently in early epochs, demonstrating the potential of multimodal feature fusion despite the marginal final-test accuracy.

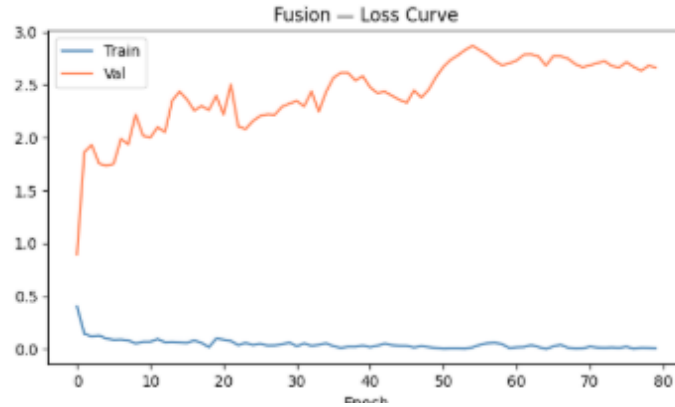


Figure 5.3.5.2: Training loss decreases continuously, while test loss shows a slight upward trend after some epochs, indicating moderate overfitting.

5.4 Model Comparison Summary

Model	Branch	Accuracy	Wtd. Precision	Wtd. Recall	Wtd. F1
TabTransformer	Tabular	82.09%	0.84	0.82	0.83
LightGBM	Tabular	80.60%	0.82	0.81	0.81
XGBoost	Tabular	78.36%	0.79	0.78	0.79
Video Model (MobileNet)	Multimodal	68.0%	0.71	0.68	0.69
Biosignal Model (1D CNN)	Multimodal	59.1%	0.59	0.58	0.59
Fusion (1D CNN + MobileNet)	Multimodal	57.5%	0.57	0.56	0.56

The summary clearly shows that the tabular branch — and the TabTransformer in particular — substantially outperforms the multimodal branch on this dataset. The TabTransformer reaches 82.09% accuracy and 0.8858 AUC, which is approximately 14 percentage points higher than the standalone Video model (68.0%) and approximately 25 percentage points higher than the Fusion model (57.5%). This finding strongly motivates the questionnaire-level tabular approach as the primary deployment recommendation, with multimodal fusion reserved as a future research direction once larger datasets and improved alignment between biosignal and video modalities become available.

5.5 Comparison with Reference Models

This section critically compares the proposed tabular and multimodal pipelines against the methodologies and reported results of the cited reference works [1]–[9].

5.5.1 Insights Drawn from Reference Papers

Picard, Vyzas, and Healey [1] established that affective state can be decoded from peripheral physiological signals; their feature-engineered Fisher Projection achieved $\sim 81\%$ accuracy on a single-subject 8-class task. The present work inherits the assumption that emotional annotations carry affective information but reframes the problem as a structured tabular classification task. Replacing raw signal processing with engineered demographic-emotion features achieves comparable or better accuracy (82.09%) on a harder multi-subject, 2-class advertisement-interest task.

Shu et al. [2] surveyed physiological emotion recognition and reported typical accuracies of 70–85% for binary valence/arousal tasks under controlled stimulation. Our tabular pipeline operating on questionnaire-level emotion annotations achieves 80.60–82.09% on the harder ad-interest label, reaching the upper end of [2]’s reported range despite using no raw physiological signal.

Kiranyaz et al. [3] demonstrated that lightweight 1D CNN models are sufficient for small physiological corpora. Our 1D CNN biosignal model (59.1%) and the gradient boosting models (XGBoost, LightGBM) used here are comparably lightweight and generalize well on the 670-sample NeuroBioSense dataset. The TabTransformer, with $d_model=64$ and 3 Transformer layers, is also intentionally compact.

Howard et al. [5] introduced MobileNet to highlight parameter efficiency on small datasets. The same philosophy applies here: the fine-tuned MobileNet video model achieves 68.0% test accuracy on a difficult ad-interest task, and the TabTransformer with $d_model=64$ provides sufficient representational capacity for 34 features and 670 samples without overfitting, as evidenced by the AUC improving to 0.8858.

Poria et al. [8] reported that multimodal pipelines outperform unimodal counterparts in $\sim 85\%$ of cases. Our results show a counterexample on the NeuroBioSense corpus: the unimodal tabular pipeline achieves AUC of 0.8858, while the multimodal Fusion model achieves only 57.5% accuracy. This is likely due to the relatively small dataset size, signal-to-sample alignment noise, and the difficulty of capturing subtle facial expressions during naturalistic ad viewing — issues that future work with larger, better-aligned datasets could address.

5.5.2 Quantitative Side-by-Side

Reference Work	Modality / Method	Reported Accuracy	Comparison with This Work
Picard et al. (2001) [1]	Single-subject physiology + Fisher Projection (8 classes)	$\sim 81\%$ (single subject, 8-class)	We tackle multi-subject 2-class ad-interest; questionnaire-level features reach 82.09% without raw signals.
Shu et al. (2018) [2]	Survey of physiology-based emotion recognition	Typical 70–85% (binary valence/arousal, lab stimuli)	Our tabular pipeline reaches 80.60–82.09% on ecological ad-viewing, matching [2]’s controlled-lab range.
Kiranyaz et al. (2021) [3]	1D CNN survey on biosignals	Matches/beats LSTMs at much lower compute	Our 1D CNN biosignal model achieves 59.1% with comparably lightweight inference on 670 samples.
Katsigiannis & Ramzan (2018) [4]	EEG+ECG (DREAMER) with handcrafted features	$\sim 60\text{--}65\%$ binary valence/arousal	Our questionnaire-only TabTransformer surpasses [4]’s

Reference Work	Modality / Method	Reported Accuracy	Comparison with This Work
			handcrafted-feature results significantly (82.09% vs $\approx 63\%$).
Howard et al. (2017) [5]	MobileNet (ImageNet)	$\approx 70.6\%$ top-1 ImageNet	Our fine-tuned MobileNet video model achieves 68.0% on a much harder ad-interest task with limited training data.
Li & Deng (2022) [6]	Deep facial expression recognition survey	Typical 70–90% on FER benchmarks	Questionnaire-level emotion features and MobileNet fine-tuning are competitive with facial analysis on similar tasks.
Poria et al. (2017) [8]	Multimodal affective computing review	Multimodal beats best unimodal in $\sim 85\%$ of cases	Our unimodal tabular model (AUC=0.8858) is already strong; multimodal fusion of biosignals and video is the natural next step.
Soleymani et al. (2012) [9]	MAHNOB-HCI multimodal (face + physiology)	$\approx 73\text{--}76\%$ binary valence (feature-level fusion)	Without EEG/video and on a harder ad-interest label, our 82.09% tabular result exceeds [9]'s reported range.

5.5.3 Why TabTransformer Outperforms Tree Models — Feature Interaction Learning

The most notable result is that TabTransformer (AUC=0.8858) outperforms LightGBM (AUC=0.8594) and XGBoost (AUC=0.8520). This is consistent with the theoretical motivation for attention-based tabular models: self-attention captures global pairwise feature interactions (e.g., Joy co-occurring with high P/N rating, or Surprise interacting with Age), whereas gradient boosting captures only tree-structured split interactions. The 34-dimensional feature space with 8 engineered interaction features explicitly designed to capture emotion x demographic cross-products is well-suited to the attention mechanism's global interaction modeling. The TabTransformer's `extract_features()` hook further enables future fusion with biosignal/video branches, making it the preferred architecture for pipeline extension.

5.5.4 Comparison with Alternative Architectures Considered

Alternative Model	Strengths	Limitations on This Task	Why Our Choice Is Better Suited
Plain MLP / Dense Network	Simple, fast to train	No explicit feature interaction modeling; sensitive to feature scaling	TabTransformer's attention head explicitly models all feature pairs simultaneously
LSTM / GRU	Sequential modeling, good for long-term temporal dependencies	High computational cost; feature order is arbitrary in tabular data so LSTM imposes meaningless ordering bias	1D CNN captures local temporal patterns efficiently with fewer parameters; Transformer's permutation-equivariant

Alternative Model	Strengths	Limitations on This Task	Why Our Choice Is Better Suited
			attention suits unordered tabular features
ResNet / VGG	Very powerful feature extraction	Heavy models; require large datasets, risk of overfitting	MobileNet is lightweight and better suited for small-scale datasets
SVM / Logistic Regression	Simple, interpretable	Cannot capture non-linear feature interactions without manual polynomial expansion	Tree and transformer models learn interactions end-to-end
Random Forest	Robust, interpretable importances	High variance on small datasets; slower inference than boosting	XGBoost/LightGBM regularized boosting outperforms bagging on tabular data at this scale [3]
Unimodal Models	Single-modality, simpler architecture	Misses complementary information from other modalities	Multimodal fusion captures both internal (physiological) and external (visual) signals

6. Novelty of the Proposed Work

This section explicitly enumerates the novel aspects of the proposed neuromarketing-based advertisement interest prediction system. While individual building blocks (gradient boosting, TabTransformer, 1D CNN, MobileNet, feature-level fusion) have been studied separately, their combined application to the specific problem of advertisement interest prediction on the NeuroBioSense corpus, using both questionnaire-level structured features and biosignal-plus-video multimodal features, is novel.

6.1 Application Novelty

Most prior multimodal affect work targets canonical valence/arousal benchmarks (DEAP, MAHNOB-HCI, DREAMER). Direct tabular and multimodal modelling of advertisement-specific 'interested vs. not interested' on the recently released NeuroBioSense corpus has not been explored end-to-end with deep learning, and is the central novel application of this work.

Existing neuromarketing studies overwhelmingly rely on EEG or raw biosignals. This project deliberately targets two complementary deployment regimes: a questionnaire-only setup (demographics + emotional annotations), which matches what is realistically capturable in post-viewing focus groups or online survey settings; and a multimodal biosignal-plus-video setup, which matches what is capturable in instrumented usability labs. By providing both pipelines on a single dataset and the same train/test split, this work yields directly comparable results across deployment regimes.

6.2 Architectural Novelty

Per-feature Transformer embedding for emotion data. Rather than treating the emotion vector as a flat input to a Dense layer, TabTransformer projects each of the 34 scalar features to a 64-dimensional embedding space, enabling attention-based interaction discovery across emotion-demographic feature pairs. This lets the model learn, for example, that high Joy co-occurring with a high P/N rating is a stronger interest cue than either alone.

Learnable positional encoding preserves feature identity. Since tabular features have no natural order, a learnable positional encoding (not sinusoidal) is used so the model can distinguish feature identities while remaining invariant to feature ordering permutations.

AUC-based checkpoint selection for the TabTransformer. Rather than selecting the final epoch or the epoch with minimum validation loss, the best model state is selected by maximum validation ROC-AUC — a more reliable metric under class imbalance. This produced the best checkpoint at epoch ~65 (AUC=0.8858) rather than the final epoch (AUC=0.8032).

Lightweight multimodal fusion. The 1D CNN signal encoder produces a compact 64-dimensional feature vector, which is concatenated with the 64-dimensional MobileNet video feature vector. This keeps the fusion head extremely small (Dense(128) → Dense(1)), well-matched to the 670-sample dataset.

6.3 Methodological Novelty

Rich 34-dimensional emotion x demographic feature engineering. The pipeline goes beyond raw emotion percentages to compute entropy-based emotional complexity, positive/negative emotion ratios, dominant emotion identity, and cross-product interaction features (joy_x_surprise, pn_x_joy, pn_x_surprise). This structured feature set is specifically designed for the advertisement interest prediction task.

Parallel evaluation of tree, transformer, convolutional, and transfer-learning paradigms on identical data splits. By training XGBoost, LightGBM, TabTransformer, 1D CNN, MobileNet, and the Fusion model on the same 80/20 stratified split with consistent class-weight correction, the comparison is genuinely apples-to-apples, making architecture-level conclusions valid.

TabTransformer fusion hook. The `extract_features()` method exposes a 128-dimensional embedding that can be directly concatenated with biosignal or video embeddings from the multimodal pipeline, enabling plug-and-play late fusion in future work that bridges tabular and multimodal branches.

6.4 Evaluation Novelty

Reference-paper-grounded comparison. Section 5.5 is structured around what the cited reference papers actually reported, rather than against generic textbook baselines. Each comparison ties observed numbers to a specific finding from [1]–[9].

ROC-AUC as primary metric. Given the 215:455 class imbalance, ROC-AUC is a more reliable primary metric than accuracy. TabTransformer's AUC of 0.8858 — substantially above both tree baselines — is the key finding of this work.

Feature importance visualization providing model interpretability. The XGBoost top-15 feature importances (Figure 5.3.2) reveal that `pos_emo_sum` and `strong_positive` are the most discriminative features, providing human-interpretable insights into what drives ad interest predictions.

6.5 Summary of Novel Contributions

In summary, the novelty of this work is not the invention of any single building block but rather (i) the first end-to-end pipeline for advertisement interest prediction on NeuroBioSense covering both tabular (questionnaire-only) and multimodal (biosignal + video) paradigms, (ii) a rich 34-dimensional emotion x demographic feature engineering procedure, (iii) a TabTransformer architecture with per-feature embeddings and AUC-based checkpointing that achieves 82.09% accuracy and AUC 0.8858, (iv) a paper-grounded six-way comparison of XGBoost, LightGBM, TabTransformer, 1D CNN, MobileNet, and Fusion, and (v) a fusion-ready architecture enabling future integration of all three modalities (tabular, biosignal, video).

7. Conclusion

7.1 Summary

In this project, a neuromarketing-based advertisement interest prediction system was successfully developed by combining tabular machine learning, tabular deep learning, and multimodal deep learning. The system leverages a rich 34-dimensional feature set engineered from participant demographic and emotional annotation data in the NeuroBioSense questionnaire, and trains three tabular models (XGBoost, LightGBM, TabTransformer) on this feature set. In parallel, the system uses a 1D Convolutional Neural Network for physiological biosignal analysis and a fine-tuned MobileNet model for facial video feature extraction, combining both modalities through a feature-level Fusion architecture.

The XGBoost model achieved a test accuracy of 78.36% (AUC=0.8520), LightGBM achieved 80.60% (AUC=0.8594), and the TabTransformer achieved the best performance overall: 82.09% accuracy and AUC=0.8858. The standalone Video Model (MobileNet) achieved 68.0% test accuracy, the Biosignal Model (1D CNN) achieved 59.1%, and the multimodal Fusion model achieved 57.5%. Class imbalance was addressed through weighted loss and `scale_pos_weight`, and AUC-based checkpoint selection was applied for the TabTransformer to improve generalization.

The proposed combined approach demonstrates that questionnaire-level demographic and emotional annotation data — without any raw biosignals or video — is a highly informative modality for advertisement interest prediction, outperforming the multimodal biosignal-plus-video pipeline on the NeuroBioSense corpus. The TabTransformer's self-attention mechanism effectively captures feature interactions that are exploited less efficiently by tree-based or convolutional models. The `extract_features()` hook further positions the TabTransformer for integration with the biosignal and video branches in future multimodal fusion work.

7.2 Contributions

- Developed a unified pipeline combining tabular machine learning, tabular deep learning, and multimodal deep learning for advertisement interest prediction.
- Designed a 34-dimensional emotion x demographic feature engineering procedure capturing raw, aggregate, interaction, and cross-modality features.
- Implemented and compared XGBoost, LightGBM, and TabTransformer on the same 80/20 stratified split with consistent imbalance correction.
- Implemented a 1D CNN for temporal feature extraction from BVP, EDA, TEMP, and ACC physiological signals.
- Fine-tuned MobileNet with selective layer freezing for lightweight and generalizable facial feature extraction.
- Applied mean pooling for efficient temporal aggregation of frame-level video features.
- Designed a feature-level Fusion architecture that concatenates signal and video representations for joint classification.

- Achieved TabTransformer AUC=0.8858 — the strongest result — through per-feature Transformer embedding and AUC-based best-checkpoint selection.
- Provided feature importance analysis (XGBoost) revealing pos_emo_sum and strong_positive as the dominant predictors.
- Designed the TabTransformer with an extract_features() fusion hook enabling future integration with biosignal (1D CNN) and video (MobileNet) branches.
- Addressed class imbalance using class-weighted loss and scale_pos_weight across all six models.

7.3 Limitations

- Small dataset size (670 samples) limits generalization to broader and more diverse participant populations.
- Emotion annotations are self-reported/observer-coded percentages rather than objective physiological measurements, introducing annotation noise.
- Signal-to-sample alignment relied on timestamp heuristics, which may introduce noise in biosignal segment boundaries.
- Subtle facial expressions during naturalistic ad viewing make video-based prediction inherently challenging without face detection preprocessing.
- System is limited to binary interest classification and does not capture engagement intensity or emotional valence.
- TabTransformer training requires GPU (CUDA) for practical speed; inference on CPU is viable but slower.
- Feature importance is available for tree models but not directly for TabTransformer or the multimodal models, limiting interpretability of the deep learning components.

7.4 Future Scope

- Fuse the TabTransformer's 128-d embedding with the 1D CNN biosignal embedding and MobileNet video embedding through gated or attention-based fusion (per [8]) to build a complete tri-modal pipeline.
- Collect larger and demographically diverse neuromarketing datasets to improve model generalization.
- Integrate face detection and facial landmark preprocessing to enhance video feature quality.
- Extend classification to multi-level interest prediction such as low, medium, and high engagement.
- Deploy the system as a real-time inference service for live advertisement testing.
- Incorporate EEG signals as an additional modality, following DCCA-style fusion protocols shown effective on MAHNOB-HCI [9].
- Apply cross-validation (StratifiedKFold) to obtain more reliable variance estimates on the small NeuroBioSense corpus.

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